Direct. Chapter-wise weightage + priority analysis for **Class 12 Biology (2025–26)** under Central Board of Secondary Education based on official pattern (70-mark theory, 30 practical).

Reference text: NCERT Biology Class 12

🔬 UNIT VI – REPRODUCTION (16 Marks)

1️⃣ Sexual Reproduction in Flowering Plants (≈6–7 marks)

Weight Level: 🔴 High
Why important: Frequently gives 5-mark descriptive + diagram question.
Question trend:

- Diagram based (embryo sac / L.S. of flower)
- Process explanation (double fertilisation)
- Short note (apomixis / polyembryony)
  Preparation depth: Line-by-line NCERT.

2️⃣ Human Reproduction (≈5–6 marks)

Weight Level: 🔴 High
Trend:

- Menstrual cycle (very common 3–5 marks)
- Spermatogenesis / Oogenesis
- Hormonal control
- Diagram of reproductive system
  Scoring tip: Flowchart answers fetch full marks.

3️⃣ Reproductive Health (≈3–4 marks)

Weight Level: 🟡 Moderate
Mostly short notes + case based.
Focus on contraception methods + STDs.

🧬 UNIT VII – GENETICS & EVOLUTION (20 Marks) 🔥 Highest Weight

4️⃣ Principles of Inheritance & Variation (≈7–8 marks)

Weight Level: 🔴 Very High
Most numerical chapter.
Trend:

- Mendelian crosses (3–4 marks problem)
- Pedigree analysis
- Sex-linked inheritance
  If you master this → easy marks.

5️⃣ Molecular Basis of Inheritance (≈7–8 marks)

Weight Level: 🔴 Very High
Often gives long answer.
Trend:

- DNA replication
- Transcription / Translation
- Lac operon
- DNA structure diagram
  Concept heavy but scoring.

6️⃣ Evolution (≈4–5 marks)

Weight Level: 🟡 Moderate
Mostly theory based:

- Hardy Weinberg
- Evidences of evolution
  Short explanations.

🏥 UNIT VIII – BIOLOGY & HUMAN WELFARE (12 Marks)

7️⃣ Human Health & Disease (≈5–6 marks)

Weight Level: 🔴 High
Immunity questions very common.
Case study possible.

8️⃣ Microbes in Human Welfare (≈4–5 marks)

Weight Level: 🟡 Moderate
Industrial applications + sewage treatment.

🧪 UNIT IX – BIOTECHNOLOGY (12 Marks)

9️⃣ Biotechnology: Principles & Processes (≈6–7 marks)

Weight Level: 🔴 High
PCR + cloning steps almost fixed question.
Diagram expected.

🔟 Biotechnology & Its Applications (≈5–6 marks)

Weight Level: 🟡 Moderate–High
GM crops / insulin production / ethics.

🌍 UNIT X – ECOLOGY (10 Marks)

1️⃣1️⃣ Organisms & Populations (≈2–3 marks)

Low weight but easy theory.

1️⃣2️⃣ Ecosystem (≈3–4 marks)

Energy flow diagram frequently asked.

1️⃣3️⃣ Biodiversity & Conservation (≈2–3 marks)

Hotspots + conservation methods.

1️⃣4️⃣ Environmental Issues (≈2–3 marks)

Pollution + ozone depletion.

📊 Priority Ranking (Most Important → Least)

1. Molecular Basis of Inheritance
2. Principles of Inheritance
3. Sexual Reproduction in Flowering Plants
4. Human Reproduction
5. Biotechnology Principles

6. Human Health & Disease
7. Biotechnology Applications
8. Evolution
9. Microbes in Human Welfare
10. Ecology chapters

🎯 Strategy Allocation (Time %)

- Genetics Unit → 30% preparation time
- Reproduction → 25%
- Biotechnology → 20%
- Human Welfare → 15%
- Ecology → 10%